

Program Structure



General:

Live Teaching - Live Teaching sessions from Monday – Saturday from 08:45 AM to 05:45 PM.

For the Graded Assessments (quiz and project), the submission timeline will be

Graded Project- 2 hours

Graded Quiz- 45 minute

Day	Day	Day	Modules	Sub Topics
Day 1	04 May 2026	Monday	Java Language Foundation	Java Basics + Frontend
				Introduction to Java and Loops
				Conditional Statements
				Operators
				Data Types
				Wrapper Classes
				Arrays
				Strings
				String Buffer
				String Builder
Day 2	05 May 2026	Tuesday	OOPS Properties	Regular Expressions
				Static and non static Members
				Memory Allocation for Static and Non Static member
				Abstraction & Interfaces
				Encapsulation
Day 3	06 May 2026	Wednesday	HTML and CSS	Polymorphism
				Inheritance, super keyword and super method
				Basic Structure of an HTML Page
				Various Tags and Attributes
				Inline and Block Elements
				Forms
				Semantic elements
				HTML 5 Features
				Paper based Layout
				Hero unit
				Audio Video Players
				Semantic Markup
				Geo Location
				Intro to CSS
				CSS Properties and their syntaxes
				Various Selectors in CSS and their preference orders
				Inline , Internal and External Styling
				Box Model
				CSS 3
Day 4	07 May 2026	Thursday	BootStrap	Advance Unites like rem, em and vh vw
				New elements added like Round Corners
				Working with flex box
				Working with BootStrap
				Cards, Tables, and Lists
				Develop a static webpage
			Java Script	Forms and Form Validations
				Navs and Navbar
				Pagination and Progress
				Navigation and Pagination
				Comparing material UI in Angular with Bootstrap based on demo
				Basics of JavaScript - Data type, variables, Type Conversions, Basic Operators, branching and Looping Statements, Functions, Arrays and Strings
Day 5	08 May 2026	Friday	Fetch APIs	Modern JavaScript (ES6) - ES6 Features, "Let" keyword, "Const" Key word, "Spread" Operator, Lamda Function, Classes, Map & Set Object, Promises, "ForEach", Deafult Parameters, Javascript Modules, Asynchronous Java script, Synchronous Vs Asynchronous Javascript, Call Backs, Async & Await
				JSON and Restful APIs using "Async & Await"
				JavaScript - Fetch API
				DOM Manipulation
				Error Handling using JS
				Java Script Design Patterns (Introduction)
				Typescript + React Basics
				Var-hoisting
				Functional Score
				Classes & Methods & Constructor
				Maps

Day 6	09 May 2026	Saturday	TypeScript	Iterators
				Interfaces
				Declarations and Annotations
				Anytype
				Enumeration
				Decorator
				Arrays & tuples
Day 7	11 May 2026	Monday	React Basics	1. React Overview, Virtual DOM concepts, JSX concepts create-react-app (React Project generation and Build Tool)
				2. React Components (Class, Functional), Properties, Event Handling
				3. React State, Controlled Components
				4. Using refs (Uncontrolled Components)
				5. PropTypes
				6. Component Life Cycle
				7. CSS Styling, Bootstrap in React
				8. Composable Components (Multiple Components)
Day 8	12 May 2026	Tuesday		React application interfacing with backend server for data (json-server)
				9. React Router
				10. Routing Transition Concepts (Transition into and Transition from)
Day 9	13 May 2026	Wednesday	Milestone - 1	
Day 10	14 May 2026	Thursday	React Basics	12. Form handling using Formik, Yup
				13. Flux Pattern (Action, Dispatcher, Stores)
				Dependency Injection
				Component Life cycle and basic building blocks of SPA Framework
Day 11	15 May 2026	Friday	Advance React + Database(SQL+ MongoDB) - 5 Days	
			ReactJs	1. Lazy Loading, Code Splitting
				2. Pure Components
				3. Error Boundary
				4. Portals
Day 12	16 May 2026	Saturday	React Hooks	5. React Context API
				6. Progressive Web App with React
				7. React Hooks
				8. State management - Redux, nxrx & nxrx-effects
Day 13	18 May 2026	Monday	SQL	Introduction to MySQL
				Basic Select and CRUD operations
				DDL
				DML
				DQL
				TCL
				Normalization
				Joins
				Subquery
Day 14	19 May 2026	Tuesday		Aggregate Functions
				Date function
				Query optimization Basics
				Indexes
				Query optimization using Indexes
Day 15	20 May 2026	Wednesday	Mongodb	Introduction to NoSQL
				Introduction to MongoDB
				MongoDB CRUD Operations
				MongoDB Data Modeling
				Indexing and Query Optimization
				Introduction to MongoDB Atlas
				Aggregation Framework in MongoDB
Day 16	21 May 2026	Thursday	Node JS and Express JS frameworks	
			Introduction to Node.js	What is Node.js?
				History of Node.js
				Features and advantages
			Node.js Basics	Installing Node.js
				Understanding npm (Node Package Manager)
				Creating a simple Node.js application
Day 17	22 May 2026	Friday	Node.js Core Modules	fs (File System)
				http/https (HTTP/HTTPS Servers and Clients)
				path (File and Directory Paths)
				events (EventEmitter)
			Asynchronous Programming in Node.js	Callbacks
				Promises
				Async/Await
				What is Express.js?
			Introduction to Express.js	Why use Express.js?
				Features and advantages
				Getting started with Express.js

Day 18	23 May 2026	Saturday	Setting Up Environment	Installing Node.js and npm (Node Package Manager)
				Creating a new Node.js project
				Installing Express.js in your project
			Basic Routing	Understanding routes in Express.js
				Creating routes for different HTTP methods (GET, POST, PUT, DELETE)
				Handling dynamic routes and parameters
				Using route middleware
Day 19	25 May 2026	Monday	Middleware, Authentication, API, Testing & Deployment	
			Middleware	What is middleware in Express.js?
				Writing custom middleware functions
				Using built-in middleware (e.g., bodyParser, morgan)
			Working with Templates	Error handling middleware
				Introduction to template engines (e.g., Pug, EJS, Handlebars)
Setting up template engines in Express.js				
				Rendering dynamic content using templates
Day 20	26 May 2026	Tuesday	Working with Forms and Data	Handling form submission
				Parsing form data with bodyParser middleware
				Validating form data
			JWT Authentication and Authorization	Interacting with databases (e.g., MongoDB, MySQL) using middleware (e.g., Mongoose, Sequelize)
				Implementing user authentication
				Using middleware for authentication (e.g., Passport.js)
	Managing user sessions and cookies			
				Role-based access control (RBAC)
Day 21	28 May 2026	Thursday	Milestone -2	
Day 22	29 May 2026	Friday	RESTful APIs	What is RESTful architecture?
				Creating RESTful APIs with Express.js
				Handling CRUD operations (Create, Read, Update, Delete)
			Middleware for RESTful APIs	Consuming APIs using AJAX or Fetch API
				Using middleware for input validation (e.g., Joi, Express-validator)
				Handling errors in APIs
Day 23	30 May 2026	Saturday	File Upload and Download	Implementing rate limiting and throttling
				Handling file uploads with Express.js
				Using middleware for file upload (e.g., multer)
			Real-time Communication	Serving static files
				Introduction to WebSocket and Socket.io
				Setting up WebSocket server with Express.js
Day 24	01 June 2026	Monday	Testing Express.js Applications	Implementing real-time features (e.g., chat applications)
				Writing unit tests with tools like Mocha, Chai
				Testing routes and middleware
			Deployment and Production	Integration testing with SuperTest
				Deploying Express.js applications to hosting platforms (e.g., Heroku, AWS)
				Configuring production environment
Day 25	02 June 2026	Tuesday	Backend + Frontend Integration	
			Database Connectivity (SQL)	Performance optimization techniques
				Overview of backend–database integration, setting up MySQL with Node.js, using MySQL drivers (mysql2), creating database connections and connection pools, performing CRUD operations from Node.js, error handling and best practices
				NoSQL Connectivity
			Connecting Node.js to MongoDB using Mongoose Schema & Model creation	
			CRUD operations with MongoDB	
Day 26	03 June 2026	Wednesday	NoSQL Connectivity	Introduction to ORM/ODM concepts
				Using Sequelize ORM – overview & comparison
				Capstone Project induction
			Real-time Communication	Introduction to Socket.IO
				Client–server communication model
				Setting up Socket.IO with Node.js & Express
Day 27	04 June 2026	Thursday	Real-time Communication	Broadcasting & rooms
				Real-time use cases (chat, notifications)
				Error handling & scalability considerations
			Full Stack Application – Part 1	Application architecture (Frontend + Backend + DB)
				Requirement walkthrough & user stories
				Backend API design & integration
Day 28	05 June 2026	Friday	Full Stack Application – Part 1	Frontend consumption of APIs
				Authentication & session flow
			Full Stack Application – Part 2	

Day 28	05 June 2026	Friday	Full Stack Application – Part 2	End-to-end data flow implementation Integrating real-time features (Socket.IO) Validation, error handling & logging Testing & debugging Deployment-ready configuration Code review & best practices
Day 29	06 June 2026	Saturday	Generative AI	AI Foundations: Introduction to AI, Evolution of AI, AI vs ML vs GenAI, Key concepts – Supervised Learning, Unsupervised Learning, AI in daily life, AI in IT Services & Enterprise, Industry use cases across SDLC Generative AI & Tools: What is Generative AI, LLM basics, Text/Code/Image generation, Overview of GenAI tools, Introduction to Microsoft Copilot & Developer Copilots, Prompt engineering basics, Demo examples AI for Daily Work Productivity: Content & email generation, Documentation & summarization, Code assistance using Copilot, Presentation creation, Learning & research acceleration, Meeting notes automation, Developer copilots in SDLC Responsible AI & Future of Work: Ethical AI principles, Bias & fairness, Data privacy & security, Responsible usage guidelines for GenAI & Copilot, AI-augmented roles, Future skills & career readiness, Knowledge check & Q&A
Day 30	08 June 2026	Monday	Final Milestone	
Day 31-35	09 June 2026 - 12 June 2026	Tuesday - Friday	Capstone Project presentation	